

Quant Derivatives — Interview Theory Guide

Deep-Dive Companion to the Theory Pack

Full derivations, intuition, and worked interview Q&A for every topic — probability foundations through Black–Scholes, Greeks, Monte Carlo, implied volatility, delta-hedging, calibration, and model validation.

Each chapter follows the same pattern: **core theory** → **derivation / mechanics** → **intuition** → **interview Q&A**. Read a chapter, close the PDF, and try to reproduce the derivation and answer the Q&A from memory before moving on.

Chapter 0 — Prerequisites in Interview Depth

You won't be examined on prerequisites directly, but they surface constantly inside derivative-pricing questions. Know these cold enough to use them mid-derivation without breaking stride.

Probability essentials

A random variable's expectation $E[X]$ is its probability-weighted average outcome; variance $\text{Var}(X) = E[X^2] - (E[X])^2$ measures spread. Covariance $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ measures co-movement. Conditional expectation $E[X|Y]$ is the best predictor of X given knowledge of Y — this concept underlies risk-neutral pricing, where a derivative's price is a conditional expectation of its discounted payoff.

The Normal distribution is symmetric and fully described by mean and variance. The Lognormal distribution arises when the log of a variable is Normal — this is exactly the assumption Black–Scholes makes about stock prices, because GBM makes $\ln(S_T)$ Normal.

The Central Limit Theorem (CLT) says that averages of many i.i.d. random variables approach a Normal distribution regardless of the underlying distribution — this is the theoretical engine behind Monte Carlo confidence intervals.

Statistics essentials

Standard error is the standard deviation of an estimator itself (not of the data) — it shrinks as sample size grows, specifically as $1/\sqrt{n}$. This single fact drives the entire Monte Carlo convergence discussion in Chapter 17.

Calculus essentials

Partial derivatives (treating other variables as constant) are literally what the Greeks are. Taylor expansion — approximating a function via its derivatives at a point — is the mechanical tool used to derive Itô's Lemma. The chain rule extends naturally to Itô's chain rule once you add the second-order term.

Q: Why does Black–Scholes assume stock prices are lognormal rather than normal?

A: Because a Normal distribution allows negative prices, which is economically meaningless for a stock. Modeling returns (or log-price) as Normal instead keeps the price itself always positive, while still giving tractable, closed-form mathematics.

Chapter 1 — Financial Foundations: Forwards

What a derivative is, precisely

A derivative's value is defined entirely by another asset's value (the 'underlying'), not by any intrinsic value of its own. This single fact is why every pricing technique in this guide reduces to two ideas: figure out the payoff as a function of the underlying, then find a portfolio of tradeable instruments that replicates that payoff.

Forward pricing

$$F = S \cdot e^{((r - q) \cdot T)}$$

Full derivation — cash-and-carry arbitrage

Compare two strategies that both deliver one share of stock at time T :

- Strategy A: enter a forward contract to buy the stock at T for price F .
- Strategy B: borrow $S \cdot e^{(-qT)}$ today, buy $e^{(-qT)}$ shares now, reinvest all dividends into more shares so you hold exactly 1 share at T , then repay the loan (which has grown to $S \cdot e^{(-qT)} \cdot e^{(rT)} = S \cdot e^{((r-q)T)}$).

Both strategies deliver exactly one share at T with zero net cash flow in between. By the law of one price they must cost the same today, so the forward price must equal the loan repayment: $F = S \cdot e^{(r-q)T}$. If F were higher, you'd sell the forward and do Strategy B (buy low via borrowing, deliver into the expensive forward) for a riskless profit; if F were lower, you'd do the reverse. Either mispricing is arbitrated away instantly by rational traders — which is exactly why the formula must hold in an efficient, frictionless market.

Q: If a stock pays no dividends, is the forward price above or below the spot price (assuming $r > 0$), and why?

A: Above. With $q = 0$, $F = S \cdot e^{rT} > S$, because the seller of the forward, absent dividends, only forgoes the time value of money by deferring delivery — you're effectively financing the purchase, and that financing cost is the risk-free rate.

Q: What happens to the forward price argument if borrowing and lending rates differ?

A: The no-arbitrage bound becomes a range rather than a single price: F must sit between $S \cdot e^{(r_{\text{borrow}} - q)T}$ and $S \cdot e^{(r_{\text{lend}} - q)T}$, because you'd use the borrowing rate to exploit an overpriced forward and the lending rate to exploit an underpriced one, and the two directions face different funding costs.

Chapter 2 — Options Theory

Payoffs, and why they're kinked

Call payoff = $\max(S_T - K, 0)$ Put payoff = $\max(K - S_T, 0)$

The 'max' function is what makes options nonlinear — this single kink is the entire reason Gamma exists (linear instruments like forwards have zero Gamma, because their payoff is a straight line with no curvature).

Intrinsic value vs. time value

Intrinsic value is what the option would be worth if exercised right now: $\max(S - K, 0)$ for a call. Time value is the remainder — the market's price for the possibility that the option becomes more valuable before expiry. Time value is maximized ATM and decays to zero at expiry (Theta decay), which is why ATM options have the richest optionality and the fastest decay near expiration.

Moneyness

Moneyness = S / K Log-moneyness = $\ln(S / K)$

Log-moneyness is preferred in quant work because it's symmetric: a call 10% ITM and a put 10% ITM have log-moneyness values that are exact negatives of each other, which isn't true of the raw ratio.

Q: Why is an ATM option's Gamma the highest?

A: Gamma measures how fast Delta changes as the stock moves. ATM is exactly where a small stock move is most likely to flip the option's fate between finishing in- or out-of-the-money, so Delta is most sensitive to the stock price there. Deep ITM or OTM, the outcome is already nearly certain, so a small stock move barely changes Delta.

Chapter 3 — Put-Call Parity

$C - P = S \cdot e^{(-qT)} - K \cdot e^{(-rT)}$

Derivation via replication

Consider two portfolios at time T . Portfolio 1: long a call, short a put (both strike K). Its payoff is $\max(S_T - K, 0) - \max(K - S_T, 0)$, which algebraically simplifies to $S_T - K$ in every single state of the world — whether the option finishes above or below K . Portfolio 2: long e^{-qT} shares (reinvesting dividends to own exactly 1 share at T) financed partly by borrowing $K \cdot e^{-rT}$ (which grows to exactly K at T). Its payoff at T is $S_T - K$ as well.

Since both portfolios have identical payoffs in every future state, the law of one price says they must have identical prices today: $C - P = S \cdot e^{-qT} - K \cdot e^{-rT}$.

Practical uses

- Arbitrage detection: if quoted C , P , S violate the identity beyond transaction costs, a riskless trade exists.
- Deriving the put price from the call price (or vice versa) without a second model run.
- Engine self-test: price a call and put independently through Black–Scholes / binomial / Monte Carlo and confirm the identity holds to numerical precision — a bug that misprices one side will typically break this check immediately.

Q: What does put-call parity say about the Delta of a call and a put at the same strike?

A: Differentiating both sides with respect to S gives $\Delta_{\text{call}} - \Delta_{\text{put}} = e^{-qT}$, a constant. So the two Deltas move in lockstep — if the call's Delta rises by 0.05, the put's Delta must also rise by 0.05 (put Deltas are negative, becoming less negative).

Q: Does put-call parity hold for American options?

A: Only as an inequality, not an equality, because American options carry early-exercise value that breaks the exact replication argument (you might optimally exercise the put early, which changes its cash flow timing). For American options you instead get a bounded range: $S \cdot e^{-qT} - K \leq C_{\text{Am}} - P_{\text{Am}} \leq S - K \cdot e^{-rT}$, roughly.

Chapter 4 — No-Arbitrage Pricing

Law of one price, stated rigorously

If Portfolio A and Portfolio B produce identical cash flows in every possible future state of the world, then $\text{Price}(A) = \text{Price}(B)$ today. If not, buying the cheaper one and selling the more expensive one locks in a riskless profit today with zero net future obligation — a free lunch that competitive markets do not allow to persist.

Replication — why it's the master technique

Almost the entire course is one repeated move: find a portfolio of things you can actually trade (stock, bond, other options) whose payoff exactly matches the derivative you're trying to price, then set the derivative's price equal to the replicating portfolio's price. The binomial model, Black–Scholes' PDE derivation, and put-call parity are all instances of this single idea applied in different settings.

Q: Give a one-sentence definition of arbitrage.

A: A trading strategy with zero initial cost, no possibility of a loss, and a positive probability of profit.

Q: Why is 'no-arbitrage' a weaker assumption than 'markets are efficient' or 'investors are rational', and why does that matter for pricing?

A: No-arbitrage only requires that a handful of well-capitalized players are willing and able to exploit obvious mispricings — it says nothing about whether the price fully reflects information or whether typical investors behave rationally. That's precisely why derivative prices can be pinned down by replication without needing to model or agree on investor risk preferences or beliefs at all.

Chapter 5 — The Binomial Model, Fully Derived

Assume over one period the stock moves from S to either $S_u = S \cdot u$ (up) or $S_d = S \cdot d$ (down), with $u > 1 > d$. The option is worth C_u in the up state and C_d in the down state.

Building the replicating portfolio

Hold Delta shares of stock and B in the risk-free bond. Choose Delta and B so this portfolio matches the option's payoff in both states:

$$\text{Delta} \cdot S_u + B \cdot (1+r) = C_u \quad \text{Delta} \cdot S_d + B \cdot (1+r) = C_d$$

Subtracting the second equation from the first eliminates B:

$$\text{Delta} \cdot (S_u - S_d) = C_u - C_d \Rightarrow \text{Delta} = (C_u - C_d) / (S_u - S_d)$$

Substitute Delta back into either equation to solve for B, then the option price today is simply the cost of building this replicating portfolio: $C = \text{Delta} \cdot S + B$.

Choosing u and d

A common convention (Cox-Ross-Rubinstein) sets $u = e^{(\sigma \cdot \sqrt{dt})}$ and $d = 1/u$, chosen so the tree's variance over one step matches the stock's actual volatility over that step — this is exactly what makes the tree converge to Black–Scholes as steps increase (Chapter 7).

Why it matters: This derivation is why Delta is called the hedge ratio rather than 'the partial derivative of price with respect to spot': historically it was discovered by solving a system of two equations for a replicating trade, not by differentiating a formula. The calculus definition (dV/dS) comes later, once the model becomes continuous.

Q: Why does the real-world probability of the stock going up never appear in the binomial pricing formula?

A: Because the price is pinned down entirely by replication and no-arbitrage — two portfolios with identical future payoffs must cost the same today, regardless of how likely each future state actually is. The true probability affects how much an investor might want to hold the stock, but not the no-arbitrage price of a derivative on it.

Q: What happens to Delta as you move from a very short-dated, deep OTM call to a very short-dated, deep ITM call, in the binomial framework?

A: Delta moves from near 0 (the option almost never pays off, so the replicating portfolio needs almost no stock) toward near 1 (the option behaves almost exactly like owning the stock outright, since it's virtually certain to be exercised).

Chapter 6 — Risk-Neutral Pricing

Deriving the risk-neutral probability

Starting from the binomial replication result, algebra shows that the option's price can be rewritten as a discounted expectation using a specific probability p :

$$p = (e^{(rT)} - d) / (u - d) \quad C = e^{(-rT)} \cdot [p \cdot C_u + (1-p) \cdot C_d]$$

p is defined purely so that the stock's expected return under this measure equals the risk-free rate: $p \cdot S_u + (1-p) \cdot S_d = S \cdot e^{(rT)}$. It is a mathematical construct that falls out of the replication algebra — not a statement about what investors actually believe will happen.

The deeper idea: change of measure

Under the real-world probability measure, different investors assign different expected returns to a risky stock depending on their risk aversion, which would make derivative pricing investor-specific and untractable. Risk-neutral pricing sidesteps this entirely: because the derivative's price is fixed by replication regardless of anyone's beliefs, you're free to price it as if the world were risk-neutral — i.e., as if every asset's expected return were r — because you're not actually claiming the world is risk-neutral, only that the replication argument makes the true probabilities irrelevant to the price.

Formally, prices become the discounted expectation of the payoff under the risk-neutral measure Q :

$$\text{Price}_0 = E^Q[e^{-(rT)} \cdot \text{Payoff}_T]$$

Why it matters: This is arguably the single most important reframing in the whole course: it converts a hard economics problem (what do investors believe and how much risk premium do they demand?) into a pure mathematics problem (compute an expectation under a specific, calculable measure). Monte Carlo pricing in Chapter 16 is a direct, literal implementation of this formula.

Q: In your own words: what does 'risk-neutral measure' mean, and what does it NOT mean?

A: It means a mathematical probability measure under which every asset's expected growth rate equals the risk-free rate, used purely as a pricing device via replication. It does NOT mean real investors are indifferent to risk, and it does NOT mean the risk-neutral probabilities equal the true, real-world probabilities of the stock going up or down.

Q: Why can you replace the real-world drift μ with the risk-free rate r when simulating stock paths for Monte Carlo option pricing?

A: Because option pricing only requires the risk-neutral expectation of the discounted payoff, and under the risk-neutral measure every asset (including the stock) grows at r by construction. Using the true drift μ instead would price the option correctly only for an investor with a specific risk preference that happens to match market prices — which isn't a coherent no-arbitrage price at all.

Chapter 7 — Binomial → Black–Scholes Convergence

As the number of tree steps N grows (holding total maturity T fixed), each step's length $dt = T/N$ shrinks. With u and d chosen to match the stock's volatility over each shrinking interval, the sum of many small up/down moves behaves, in the limit, like a continuous-time process with Normally distributed log-returns — exactly the Geometric Brownian Motion assumption behind Black–Scholes. The Central Limit Theorem is doing the heavy lifting here: a large number of small, roughly-independent binary steps sums to something approximately Normal.

Practically: implement both models, price the same option across a range of N (e.g. 10, 50, 100, 500, 2000 steps), and plot binomial price vs. N . You should see the binomial price oscillate and converge toward a flat line at the Black–Scholes price as N grows.

Q: Why does the binomial price often oscillate around the Black–Scholes price rather than approaching it monotonically?

A: Because whether a given strike falls exactly on a tree node or between two nodes changes with N , creating a discretization pattern (particularly visible for American or barrier-style payoffs) that damps out as N grows large but doesn't vanish smoothly step by step.

Chapter 8 — Brownian Motion

A standard Brownian motion (Wiener process) W_t is the continuous-time limit of a random walk. It's defined by four properties, each of which matters downstream:

- $W_0 = 0$ — a normalization, not a modeling choice.
- Independent increments — the future increment $W_t - W_s$ doesn't depend on the path taken to reach W_s . This is what lets you treat each small time-step in a simulation as an independent random draw.
- Stationary, Normal increments — $W_t - W_s \sim \text{Normal}(0, t-s)$. The variance of an increment depends only on the elapsed time, not on when it occurs.
- Continuous paths — no jumps. (Real markets do jump; jump-diffusion models like Merton's relax this assumption.)

The property that breaks ordinary calculus

$(dW)^2 = dt$ (in the mean-square / quadratic-variation sense)

In ordinary calculus, $(dx)^2$ is treated as negligible compared to dx — that's what lets you drop higher-order terms in a Taylor expansion. For Brownian motion, the increment's variance over a tiny interval dt is itself dt , so $(dW)^2$ is not negligible — it's the same order as dt . This single fact is the entire reason Itô's Lemma has an extra term that ordinary calculus's chain rule doesn't.

Q: Why can't you use ordinary calculus (the regular chain rule) to differentiate a function of a Brownian-driven process?

A: Because Brownian motion has nonzero quadratic variation — $(dW)^2 = dt$ rather than 0 — so terms that would vanish in ordinary calculus survive and must be kept. Itô's Lemma is exactly the chain rule with this extra second-order correction added in.

Chapter 9 — Geometric Brownian Motion

$$dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW$$

Dividing through by S turns this into a statement about returns: $dS/S = \mu \cdot dt + \sigma \cdot dW$ — the instantaneous percentage return has a deterministic drift μ and a random shock scaled by volatility σ . Modeling in percentage (proportional) terms rather than absolute dollar terms is what keeps S strictly positive: a $-\$200$ shock could send a $\$100$ stock negative, but a -50% shock cannot.

The closed-form solution

Applying Itô's Lemma (Chapter 10) to $\ln(S)$ and integrating yields the explicit solution used directly in Monte Carlo simulation:

$$S_T = S_0 \cdot \exp\left((\mu - 0.5 \cdot \sigma^2) \cdot T + \sigma \cdot \sqrt{T} \cdot Z\right), \quad Z \sim \text{Normal}(0,1)$$

Notice the drift term inside the exponential is $(\mu - 0.5 \cdot \sigma^2)$, not simply μ — the $-0.5 \cdot \sigma^2$ correction is a direct consequence of Itô's Lemma and is often called the 'Itô correction' or 'convexity adjustment.' It exists because $E[e^X] \neq e^{(E[X])}$ for a random X — Jensen's inequality — so a naive drift would overstate the expected stock price.

Q: If $\sigma = 0$ (a deterministic stock), what does the GBM solution reduce to, and does that make sense?

A: $S_T = S_0 \cdot e^{(\mu \cdot T)}$ — ordinary continuous compounding at rate μ , with no $-0.5 \cdot \sigma^2$ correction since it's zero. This makes sense: with no randomness there's no Jensen's-inequality gap to correct for.

Chapter 10 — Itô's Lemma, Derived

Goal: find the stochastic differential of $f(S,t)$ when S follows $dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW$.

Step 1 — Taylor expand f

$$df = (df/dt)dt + (df/dS)dS + 0.5 \cdot (d^2f/dS^2)(dS)^2 + \text{higher-order terms}$$

Step 2 — substitute dS and expand $(dS)^2$

$$dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW \quad (dS)^2 = \sigma^2 \cdot S^2 \cdot (dW)^2 + (\text{terms of order } dt^{1.5} \text{ or smaller, which vanish})$$

Step 3 — apply $(dW)^2 = dt$

$$(dS)^2 = \sigma^2 \cdot S^2 \cdot dt$$

Step 4 — substitute back and collect terms

$$df = (df/dt + \mu \cdot S \cdot (df/dS) + 0.5 \cdot \sigma^2 \cdot S^2 \cdot (d^2f/dS^2)) dt + \sigma \cdot S \cdot (df/dS) dW$$

Every other term in the Taylor expansion (dt^2 , $dt \cdot dW$, dW^3 , ...) is smaller than dt in the appropriate limiting sense and is dropped — this is the technical content of stochastic calculus, but the mechanical recipe above (Taylor expand, substitute dS , replace $(dW)^2$ with dt , drop everything smaller) is all you need to reproduce the result under interview pressure.

Why it matters: The $0.5 \cdot \sigma^2 \cdot S^2 \cdot (d^2f/dS^2)$ term is exactly the term that survives only because Brownian motion has nonzero quadratic variation. Set $\sigma = 0$ and Itô's Lemma collapses back to the ordinary chain rule — a good sanity check to state out loud in an interview.

Q: Derive Itô's Lemma for $f(S,t) = \ln(S)$, and use it to explain the $-0.5 \cdot \sigma^2$ drift adjustment in GBM's solution.

A: $df/dt = 0$, $df/dS = 1/S$, $d^2f/dS^2 = -1/S^2$. Substituting into Itô's Lemma: $d(\ln S) = (\mu \cdot S \cdot (1/S) + 0.5 \cdot \sigma^2 \cdot S^2 \cdot (-1/S^2))dt + \sigma \cdot S \cdot (1/S)dW = (\mu - 0.5 \cdot \sigma^2)dt + \sigma \cdot dW$. Integrating from 0 to T gives $\ln(S_T) = \ln(S_0) + (\mu - 0.5 \sigma^2)T + \sigma \cdot \sqrt{T} \cdot Z$, which exponentiates directly to the GBM solution — showing the $-0.5 \sigma^2$ term comes straight out of the second-derivative term in Itô's Lemma applied to the log function.

Chapter 11 — The Black–Scholes Derivation, Step by Step

Step 1 — apply Itô's Lemma to the option value $V(S,t)$

$$dV = (dV/dt + \mu \cdot S \cdot (dV/dS) + 0.5 \cdot \sigma^2 \cdot S^2 \cdot (d^2V/dS^2)) dt + \sigma \cdot S \cdot (dV/dS) dW$$

Step 2 — build a hedged portfolio

Hold the option and short $\Delta = dV/dS$ shares of stock. Portfolio value: $P_i = V - (dV/dS) \cdot S$. Its change over dt :

$$dP_i = dV - (dV/dS) \cdot dS$$

Substitute dV from Step 1 and $dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW$. The $\sigma \cdot S \cdot (dV/dS)dW$ term in dV exactly cancels the $(dV/dS) \cdot \sigma \cdot S \cdot dW$ term from dS — this is the whole point of choosing $\Delta = dV/dS$. What remains:

$$dP_i = (dV/dt + 0.5 \cdot \sigma^2 \cdot S^2 \cdot (d^2V/dS^2)) dt$$

Step 3 — no randomness means no-arbitrage pins the return

dP_i has no dW term left — it's purely deterministic over the instant dt , i.e. riskless. A riskless portfolio must earn exactly the risk-free rate, or an arbitrage exists (borrow at r and buy P_i if P_i 's implied return is higher; short P_i and

lend at r if lower). So:

$$dP_i = r \cdot P_i \cdot dt = r \cdot (V - (dV/dS) \cdot S) \cdot dt$$

Step 4 — equate and simplify

Setting the two expressions for dP_i equal and rearranging gives the Black–Scholes PDE:

$$dV/dt + r \cdot S \cdot (dV/dS) + 0.5 \cdot \sigma^2 \cdot S^2 \cdot (d^2V/dS^2) - r \cdot V = 0$$

Why it matters: μ appeared in Step 1 but was completely cancelled in Step 2 — it never makes it into the final PDE. This is the crux of the whole model: because you can hedge away the stock's randomness continuously, the option's price cannot depend on the stock's expected real-world return, only on how uncertain that return is (σ), the discount rate (r), and the payoff shape. Be ready to say this in one sentence in an interview — it's one of the most commonly asked 'explain Black-Scholes' follow-ups.

Q: Where exactly does μ cancel out in the Black–Scholes derivation, and why is that surprising?

A: It cancels in Step 2, when constructing the delta-hedged portfolio $P_i = V - \Delta \cdot S$: the dW terms cancel because Δ was chosen as dV/dS , and simultaneously this specific hedge ratio also happens to make the dt -terms' dependence on μ vanish algebraically once you substitute $dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW$ into $dP_i = dV - \Delta \cdot dS$. It's surprising because intuitively you'd expect a stock's expected future return to matter for pricing a claim on that stock — but continuous hedging removes the need to take a view on μ at all.

Q: What assumption in the Black–Scholes derivation is most obviously violated in real markets, and what's the practical consequence?

A: Continuous, costless rebalancing of the hedge. Real markets rebalance discretely and pay transaction costs, so the perfect cancellation in Step 2 is only approximate in practice — this residual is exactly the 'hedge error' studied via delta-hedging simulation in Chapter 24-25, and it's why Gamma (curvature) becomes a real risk rather than a purely theoretical one.

Chapter 12 — The Black–Scholes Formula

$$C = S \cdot e^{(-qT)} \cdot N(d_1) - K \cdot e^{(-rT)} \cdot N(d_2) \quad P = K \cdot e^{(-rT)} \cdot N(-d_2) - S \cdot e^{(-qT)} \cdot N(-d_1)$$

$$d_1 = [\ln(S/K) + (r - q + 0.5 \cdot \sigma^2) \cdot T] / (\sigma \cdot \sqrt{T}) \quad d_2 = d_1 - \sigma \cdot \sqrt{T}$$

What d_1 and d_2 mean

d_2 is (up to sign/measure subtleties) the number of standard deviations by which $\ln(S_T/K)$ is expected to exceed zero under the risk-neutral measure — so $N(d_2)$ is the risk-neutral probability the call finishes in-the-money. This is also exactly the probability weight used when you write the option price as a discounted risk-neutral expectation: $C = e^{(-rT)} \cdot E^Q[\max(S_T - K, 0)]$ expands into the two-term formula above precisely because the payoff is a truncated lognormal expectation.

d_1 differs from d_2 by exactly $\sigma \cdot \sqrt{T}$, and $e^{(-qT)} \cdot N(d_1)$ is precisely Δ — this is not a coincidence but falls directly out of differentiating C with respect to S (the $N(d_1)$ and $N(d_2)$ terms' derivatives partially cancel, leaving a remarkably clean result).

Sanity checks worth memorizing

- As $\sigma \rightarrow 0$: $d_1, d_2 \rightarrow \pm \infty$ depending on moneyness, and the formula collapses to the discounted intrinsic value.

- As $T \rightarrow 0$: the formula collapses to $\max(S-K, 0)$ (or $\max(K-S, 0)$ for puts) — the payoff itself, since there's no time left for anything to change.
- As $S \rightarrow \infty$: $N(d1), N(d2) \rightarrow 1$, and $C \rightarrow S \cdot e^{-qT} - K \cdot e^{-rT}$ — the call behaves like a forward, since exercise becomes a certainty.

Q: Without deriving it from scratch, explain intuitively why the Black–Scholes call formula looks like 'S times a probability minus K times a discounted probability'.

A: It's the discounted risk-neutral expectation of the payoff, $E^Q[\max(S_T - K, 0)] \cdot e^{-rT}$, split into two pieces: the expected value of S_T conditional on finishing ITM (weighted by the probability of finishing ITM, which gives the $S \cdot e^{-qT} \cdot N(d1)$ term), minus the strike K paid only in the ITM scenario (weighted by that same probability, giving the $K \cdot e^{-rT} \cdot N(d2)$ term).

Q: What's the effect of increasing the dividend yield q on the call price, and why?

A: It lowers the call price (and raises the put price), because a higher dividend yield reduces the stock's expected forward price ($F = S \cdot e^{(r-q)T}$) falls as q rises) without changing volatility — the underlying is expected to be worth relatively less by expiry, which is unambiguously bad for call holders and good for put holders.

Chapter 13–14 — Greeks: Formulas, Derivation, Relationships

Delta

$$\Delta_{\text{call}} = e^{-qT} \cdot N(d1) \quad \Delta_{\text{put}} = -e^{-qT} \cdot N(-d1)$$

Obtained by differentiating C with respect to S . A well-known algebraic trick shows the derivatives of the $N(d1)$ and $N(d2)$ terms cancel each other out, leaving the clean $e^{-qT} \cdot N(d1)$ form rather than something messier.

Gamma

$$\Gamma = e^{-qT} \cdot N'(d1) / (S \cdot \sigma \cdot \sqrt{T})$$

$N'(x)$ is the standard Normal density (bell curve). Gamma is identical for calls and puts of the same strike — a direct consequence of put-call parity, since $\Delta_{\text{call}} - \Delta_{\text{put}} = e^{-qT}$ is constant in S , so differentiating again with respect to S gives $\Gamma_{\text{call}} - \Gamma_{\text{put}} = 0$.

Vega

$$\text{Vega} = S \cdot e^{-qT} \cdot N'(d1) \cdot \sqrt{T}$$

Always positive for both calls and puts, since more uncertainty about the terminal price can only help an option holder (the payoff is bounded below by zero, so extra spread is pure upside).

Theta

$$\Theta_{\text{call}} = -[S \cdot e^{-qT} \cdot N'(d1) \cdot \sigma] / (2 \cdot \sqrt{T}) - r \cdot K \cdot e^{-rT} \cdot N(d2) + q \cdot S \cdot e^{-qT} \cdot N(d1)$$

Usually negative for long options — value erodes as time passes, all else equal — though deep ITM puts on high-dividend stocks can occasionally have positive Theta.

Rho

$$\rho_{\text{call}} = K \cdot T \cdot e^{-rT} \cdot N(d2)$$

The identity linking them all — the PDE itself

The Black–Scholes PDE, rewritten in Greek notation, is a Greek identity you can quote directly:

$$\Theta + r \cdot S \cdot \Delta + 0.5 \cdot \sigma^2 \cdot S^2 \cdot \Gamma = r \cdot V$$

Why it matters: This means Theta and Gamma are locked in a trade-off for a delta-hedged position: high positive Gamma (common near expiry, ATM) is necessarily paired with strongly negative Theta. This is the mathematical basis of the trader's saying 'Gamma and Theta are two sides of the same coin' — you pay time decay for convexity.

Q: Why is Vega always positive for a plain vanilla long option, but Delta can be positive or negative?

A: Vega reflects sensitivity to uncertainty, and an option's payoff is bounded below (you can never lose more than the premium), so any increase in uncertainty can only increase or leave unchanged the expected payoff — never decrease it. Delta reflects sensitivity to direction, and calls profit from up-moves (positive Delta) while puts profit from down-moves (negative Delta), so its sign depends on which side of the trade you're on.

Q: Explain the Gamma/Theta trade-off in a delta-hedged option book in one or two sentences, as you'd say it in an interview.

A: For a delta-hedged position, the Black–Scholes PDE says Theta and Gamma must offset each other against the risk-free carry — high Gamma exposure (upside from convexity if the stock moves) is paid for with correspondingly steep Theta decay when it doesn't move. Traders describe this as 'renting Gamma': you pay time decay every day for the optionality, win big on large moves, and lose slowly on quiet days.

Chapter 15 — Numerical Greeks

Finite-difference formulas

$$\Delta \approx [V(S+h) - V(S-h)] / (2h) \quad \Gamma \approx [V(S+h) - 2 \cdot V(S) + V(S-h)] / h^2 \quad \text{Vega} \approx [V(\sigma+h) - V(\sigma-h)] / (2h)$$

Choosing the step size h

Too large an h introduces truncation error (the finite-difference approximation itself becomes inaccurate); too small an h introduces floating-point cancellation error (subtracting two nearly equal numbers loses precision). A typical practical choice is h proportional to S (e.g. $h = 0.01 \cdot S$), balancing both error sources — this trade-off is itself a common interview talking point.

Why compute both ways

Comparing the analytical Black–Scholes Greek against its finite-difference counterpart is a cheap, powerful validation check: any implementation bug in either the pricing formula or the Greek formula tends to show up immediately as a mismatch, without needing an external source of truth.

Q: Central differences are used for Gamma rather than a one-sided (forward) difference. Why?

A: A central difference is second-order accurate in h (error scales as h^2), while a one-sided difference is only first-order accurate (error scales as h) — for the same step size, the central difference is dramatically more precise, which matters a lot for a second-derivative quantity like Gamma that's already sensitive to noise.

Chapter 16–19 — Monte Carlo: Theory and Practice

16. The pricing principle

```
S_T = S · exp( (r - q - 0.5·sigma^2)·T + sigma·sqrt(T)·Z ), Z ~ Normal(0,1) Price ≈ e-(rT) · (1/N) · sumi( Payoff(S_T(i)) )
```

This is a direct, literal Monte Carlo estimate of the risk-neutral expectation $E^Q[e^{-(rT)} \cdot \text{Payoff}(S_T)]$ from Chapter 6 — simulate many possible risk-neutral futures, average the discounted payoff.

17. Why it converges, and how fast

By the Law of Large Numbers the sample average converges to the true expectation as $N \rightarrow \infty$. By the Central Limit Theorem, the estimator's own distribution is approximately Normal with standard error:

```
SE ≈ sample_std_dev(payoffs) / sqrt(N)
```

Why it matters: The $1/\sqrt{N}$ rate is the single most quoted fact about Monte Carlo in interviews: to halve your error, you need 4x the simulations, not 2x. This is also exactly why variance reduction techniques (below) matter — they attack `sample_std_dev` directly rather than relying purely on brute-force N .

18. Confidence intervals

```
95% CI ≈ estimate ± 1.96 · SE
```

Run the same pricing at increasing N (e.g. 1,000 / 10,000 / 100,000 / 1,000,000 paths) and show the CI width shrinking proportionally to $1/\sqrt{N}$ — a clean, quantitative convergence chart for your project write-up.

19. Variance reduction — antithetic variates

For every random draw Z , also compute the payoff using $-Z$, and average the pair. Because Z and $-Z$ are perfectly negatively correlated, the two payoffs (for many payoff shapes) are negatively correlated too, so their average has lower variance than two independent draws would — you get more precision per random number generated, effectively 'buying' variance reduction for free computationally.

Q: If a Monte Carlo price and a Black–Scholes closed-form price disagree by more than the reported 95% confidence interval would suggest is likely, what's the first thing you'd check?

A: A bug — most commonly, using the real-world drift μ instead of the risk-neutral drift $(r - q)$ when simulating S_T , an incorrect discount factor, or an off-by-one error in time (using T instead of the remaining maturity). Only after ruling out implementation bugs would you consider whether N is simply too small for the CI to be reliable.

Q: Why doesn't increasing the number of simulated time steps within a single path (as opposed to the number of paths) improve pricing accuracy for a simple European option?

A: A European option's value depends only on the terminal price S_T , which can be sampled exactly in one step using the closed-form GBM solution — simulating intermediate time steps adds no new information about S_T and only adds computational cost and discretization risk. Multiple time steps only start mattering for path-dependent payoffs (Asians, barriers, American-style exercise) where the payoff genuinely depends on the path, not just the endpoint.

Chapter 20–22 — Implied Volatility, Root-Finding, Smile

20. What implied volatility is

The σ value that, plugged into the Black–Scholes formula, reproduces an observed market price exactly. Since the formula is monotonically increasing in σ (higher Vega always positive), a unique implied vol exists for any arbitrage-free market price — but there's no algebraic way to invert the formula, so it must be solved

numerically.

21. Bisection

Start with a bracket $[\sigma_{lo}, \sigma_{hi}]$ known to contain the root (e.g. $[0.001, 5.0]$), evaluate the Black–Scholes price at the midpoint, and replace whichever endpoint has the same sign of pricing error as the midpoint. Repeat until the bracket is narrower than a tolerance.

```
sigma_mid = (sigma_lo + sigma_hi) / 2
```

Guaranteed to converge (given a valid bracket and a continuous, monotonic function) but only linearly — the error roughly halves each iteration.

21. Newton-Raphson

```
sigma_(n+1) = sigma_n - ( BS(sigma_n) - market_price ) / Vega(sigma_n)
```

Uses the local slope (Vega, which you already know how to compute analytically) to jump directly toward the root. Converges quadratically near the solution — dramatically faster than bisection when it works — but can overshoot or diverge entirely when Vega is very small, which happens for deep ITM/OTM options where the price is nearly insensitive to volatility.

Why it matters: A robust production implementation typically starts with Newton-Raphson for speed and falls back to bisection (or clips the step) if Newton produces a nonsensical negative sigma or fails to converge within a few iterations — this hybrid approach is itself a strong thing to mention in an interview.

22. Volatility smile and skew

Plotting implied vol against strike (or log-moneyness) for a fixed maturity typically shows a smile (higher implied vol for far OTM and far ITM strikes than ATM) in FX markets, or a skew (implied vol decreasing as strike rises) in equity markets — the latter reflecting that market participants pay up for downside protection (crash risk) more than for equivalent upside.

This directly contradicts Black–Scholes' core assumption of constant sigma — if the model were literally true, every strike would produce the same implied vol. The smile/skew is empirical proof the model is an approximation, motivating local volatility (Dupire), stochastic volatility (Heston), and SABR models that let volatility vary by strike and/or time and/or be itself random.

Q: Why is Newton-Raphson risky specifically for deep OTM options when solving for implied volatility?

A: Vega — the derivative used in each Newton step — approaches zero for deep OTM options because their price is barely sensitive to volatility, so the Newton update divides by a near-zero number and can produce a wildly large or even negative step, sending the next guess far outside any sensible volatility range.

Q: What does a pronounced equity index volatility skew tell you about the risk-neutral distribution of returns, relative to the lognormal distribution Black–Scholes assumes?

A: It implies the market is pricing in fatter left-tail (crash) risk than a lognormal distribution would generate — i.e., the true risk-neutral distribution has more negative skewness and/or excess kurtosis than Black–Scholes assumes, which is exactly why out-of-the-money puts trade at a higher implied vol (and are relatively more expensive) than the flat-vol model would predict.

Chapter 23–25 — Hedging, Simulation, Gamma/Hedge Error

23. Delta hedging mechanics

To hedge a long call position, hold $-\Delta$ shares of stock (i.e., sell Δ shares short). As the stock moves and Δ changes, rebalance the hedge to the new Δ . In the idealized Black–Scholes world of continuous, costless rebalancing, this hedge is perfect and the position earns exactly the risk-free rate with zero risk — this is precisely the argument used to derive the PDE in Chapter 11.

24. What a delta-hedging simulation actually does

1) Sell the option, receive the Black–Scholes premium. 2) Buy Δ shares to hedge. 3) Step forward one rebalancing period, let the stock move (simulated via GBM). 4) Recompute Δ at the new stock price and time, trade the difference in shares. 5) Repeat until expiry. 6) At expiry, settle the option payoff and tally all cash flows — the trading P&L plus the option premium received minus the payoff paid — into a final hedge P&L.

Run this thousands of times with different random paths to build a distribution of hedge P&L, then study how the distribution's spread changes with:

- Hedge frequency — daily rebalancing should show a much tighter P&L; distribution than weekly or monthly, because Δ has less time to drift between rebalances.
- Realized volatility — higher realized vol relative to the vol used to price the option widens the hedge error, since the hedge is systematically 'behind' a fast-moving stock.
- Moneyness — ATM options (highest Gamma) should show the widest hedge P&L; spread for a given rebalancing frequency.

25. Why Gamma is the source of hedge error

Between rebalances, Δ is held fixed while the stock keeps moving — the hedge is only exactly correct at the instant it's set. Gamma measures how fast the true Δ is drifting away from your fixed hedge as the stock moves, so a high-Gamma position accumulates hedging error faster between rebalances. Formally, the P&L from an unhedged Gamma exposure over a small move dS is approximately $0.5 \cdot \Gamma \cdot (dS)^2$ — always the same sign as Gamma (a long-Gamma position profits from any move, hedged or not, which is exactly offset on average by Theta decay under fair pricing).

Why it matters: This connects directly back to the Greek-identity note in Chapter 13-14: a delta-hedged book's Theta decay is the fair price paid, in expectation, for the Gamma exposure it's carrying — realized hedge P&L essentially measures whether realized volatility over the hedging period came in above or below the implied volatility baked into the option's Theta.

Q: If you delta-hedge daily instead of continuously, and the stock realizes exactly the volatility that was used to price the option, what do you expect the average hedge P&L; to be, and why isn't it exactly zero on every single path?

A: On average across many paths, hedge P&L; should be close to zero (fair value), but on any individual path it typically isn't exactly zero because discrete rebalancing leaves residual, path-dependent Gamma exposure between each rebalance — this residual only averages out to roughly zero across many independent paths, not on each one, which is exactly why the hedge P&L; has a distribution (with a spread that shrinks as rebalancing frequency increases) rather than being a single deterministic number.

Q: Intuitively, why does a short-Gamma position lose money on large moves regardless of direction, if it's been delta-hedged?

A: Being short Gamma means your hedge under-adjusts as the stock moves — you're effectively buying high and selling low as you chase the rebalancing, because your Δ lags the option's true, faster-moving sensitivity. Both up and down moves trigger this same lagging, wrong-direction rebalancing cost, so the loss shows up symmetrically for large moves in either direction.

Chapter 26–27 — Calibration & Model Validation

26. Calibration, precisely defined

Calibration means choosing model parameters so the model's output prices match observed market prices as closely as possible — it's an inverse problem, the reverse of pricing. For vanilla Black–Scholes, 'calibration' is almost always just solving for implied volatility per option (Chapter 20-21), since sigma is the only free parameter and each option effectively gets its own value. For richer models (Heston, SABR, local vol) calibration means a genuine optimization — minimizing the sum of squared pricing errors across many strikes and maturities simultaneously by adjusting a handful of shared parameters.

27. Model validation — building the checks into the engine

- Mathematical identities: put-call parity should hold to numerical (floating-point) precision — this is a near-zero-cost, always-on regression test.
- Limiting cases: $\sigma \rightarrow 0$ should collapse the price to discounted intrinsic value; deep ITM should approach the forward value; $T \rightarrow 0$ should collapse to the raw payoff.
- No-arbitrage bounds: a call must sit between $\max(S \cdot e^{-qT} - K \cdot e^{-rT}, 0)$ and $S \cdot e^{-qT}$; prices outside these bounds indicate a bug, not just a modeling choice.
- Cross-model agreement: binomial (many steps), Black–Scholes closed-form, and Monte Carlo should all agree on the same option to within their respective numerical tolerances.

Why it matters: In a Model Risk context specifically, this checklist is the actual job: independent validation of a pricing model rarely means re-deriving the mathematics from scratch, but rather systematically stress-testing exactly these kinds of identities, limits, and bounds, plus benchmarking against an independent implementation.

Q: As an independent model validator (not the model's original developer), what's the first thing you'd check on a newly submitted option pricing model, before touching any market data?

A: That it satisfies its own internal mathematical identities and limiting cases — put-call parity, convergence of the numerical method to any closed-form benchmark available, and sensible behavior at the boundaries (σ to 0, T to 0, deep ITM/OTM) — because these checks require no external data, are cheap to automate, and catch the majority of implementation bugs before you even get to the harder question of whether the model's assumptions fit the market.

Chapter 28–29 — Convergence Testing & Sensitivity Analysis

28. Convergence testing

Quantify, don't just assert, numerical accuracy: plot binomial price vs. number of steps and Monte Carlo price (with confidence interval) vs. number of simulations, and show both flattening onto the closed-form Black–Scholes benchmark. Report the empirical convergence rate (e.g. confirm CI width does in fact shrink proportional to $1/\sqrt{N}$) rather than just eyeballing a chart.

29. Sensitivity analysis

Systematically vary each input in isolation and plot the resulting option value or Greek — this both builds intuition and doubles as another validation layer (a Greek surface with an unexpected sign flip or discontinuity usually flags a bug).

- $S \rightarrow$ price and Delta/Gamma surfaces across a range of spot prices
- $K \rightarrow$ how value and moneyness-sensitivity shift across strikes
- $T \rightarrow$ Theta and the erosion of time value as expiry approaches
- $\sigma \rightarrow$ Vega and how the whole value surface scales with volatility

- $r, q \rightarrow$ Rho and dividend sensitivity, generally the smallest-magnitude effects

Q: Why is a 3D Greek surface (e.g. Delta as a function of both S and T) more informative for validation than a single 2D slice?

A: A single slice can look correct while hiding a bug that only appears at certain combinations of inputs — e.g. a discretization error that only shows up near expiry for near-the-money strikes. A full surface exposes exactly where in the input space the model behaves unexpectedly, which is far closer to how a Model Risk review would actually stress-test a pricing library in practice.